ASOS E-Commerce Inventory Optimization Analysis

1. Project Background and Objective

I analysed product pricing and size availability across the ASOS catalogue to identify inventory gaps and support purchasing and replenishment decisions. Unavailable sizes can prevent customers from completing purchases, making product availability an important area for business analysis.

The objective was to compare stockout patterns across brands and identify products with the greatest price-weighted stockout exposure. Python, Pandas, Matplotlib and Seaborn supported the analysis and visualization.

2. Understanding the Dataset

The dataset, products_asos.csv, contained 30,845 loaded records across nine original columns. These are catalogue records, rather than confirmed unique products or sales transactions.

• Product identifiers: URL, product name and SKU.

• Product attributes: size availability, category, colour and description.

• Commercial and media information: price and image links.

3. Data Preparation and Cleaning Process

I prepared the data by loading the CSV while skipping malformed lines, converting prices to numeric values using errors="coerce" and removing records with missing or invalid prices. Product descriptions were converted to strings before brand extraction.

Brand extraction used description text after the word "by", followed by a mapping to standardize selected labels, including New Look, River Island, Miss Selfridge and Top Shop. The product analysis retained brands with more than five records, while the quadrant chart included only brands with more than ten records.

4. Inventory Analysis Using Python

I created stockout metrics from the comma-separated size field by converting text to lowercase, excluding empty entries and counting entries containing "out of stock" as unavailable.

Metric	Calculation and interpretation
Stockout_Count	Number of size entries marked out of stock.
Stockout_Rate	Unavailable size entries / total listed size entries.
Lost_Revenue	Product price × Stockout_Count. I used this as a revenue exposure proxy.

For example, a product priced at 50 with three unavailable size entries has an exposure score of 150. The metric assumes one price unit per unavailable entry for comparison and does not establish that three purchases were actually lost. The currency remains unverified.

The analysis addressed the following business questions:

• Which products have the largest price-weighted stockout exposure?

• Which retained brand labels appear most frequently in the catalogue?

• How do average prices and average product stockout rates differ across brands?

• Which brands sit above the selected price and stockout thresholds?

5. Product and Brand Insights

The five largest retained brand labels by record count were ASOS (4,844), Top Shop (711), New Look (511), River Island (474) and Miss Selfridge (429). These figures measure catalogue representation, not sales volume or market share.

Ranking product rows by exposure produced the following top results:

Brand / product	Price	Unavailable entries	Exposure
Barbour - Beadnell wax jacket in black	219	9	1,971
Top Shop - premium leather collared zip product*	260	7	1,820
ASOS - premium leather trench coat*	220	7	1,540
ASOS - EDITION embellished fringe product*	250	6	1,500
Top Shop - Baggy co-ord jeans in green cord	50	27	1,350

*Product names are shortened to match the available output. Exposure values represent price units rather than verified lost sales.

6. Brand Strategy Visualization

I grouped records by Brand to calculate mean price, mean product-level Stockout_Rate, total exposure and record count, then created a bubble chart. The horizontal axis shows average price, the vertical axis shows average stockout rate and larger bubbles represent greater summed exposure.

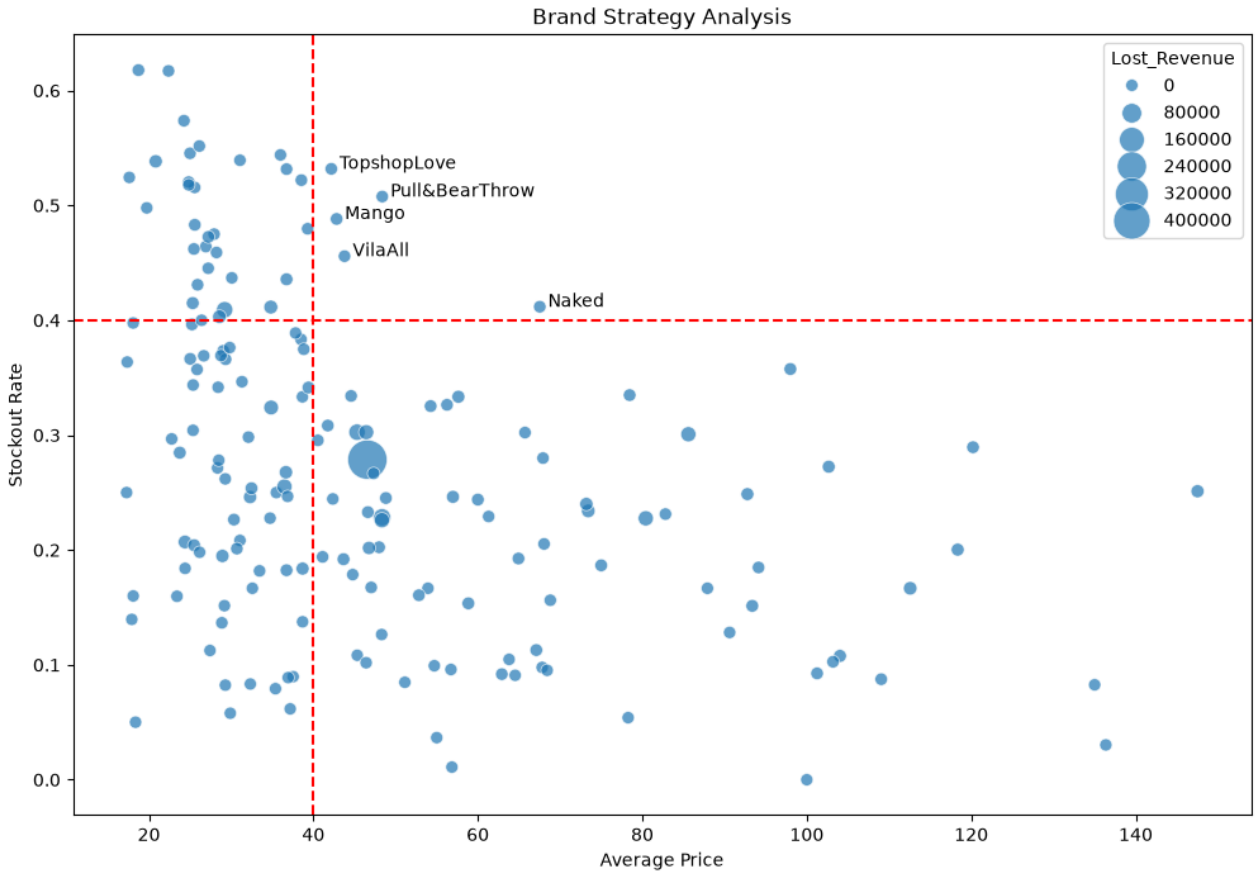


Figure 1. Brand Strategy Analysis created using Matplotlib and Seaborn.

Dashed reference lines mark price = 40 and stockout rate = 0.40, with labels identifying brands above both thresholds. These thresholds support comparison and do not represent validated measures of demand or profitability.

Position	Business interpretation
Higher price / higher stockouts	Investigate availability gaps in higher-priced products.
Lower price / higher stockouts	Review size coverage and replenishment for lower-priced products.
Higher price / lower stockouts	Check sell-through before interpreting availability as excess inventory.
Lower price / lower stockouts	Monitor availability alongside sales and inventory turnover.

7. Business Recommendations

The findings suggest the following priorities for further inventory investigation:

• Review the highest-exposure product rows first. Check the underlying size listings, current stock and replenishment dates before taking action.

• Compare brand stockout rates with actual sales, customer demand and supplier lead times to distinguish strong demand from supply constraints.

• Use size-level sales history to identify frequently requested sizes and improve replenishment planning.

• Combine price with product costs and margins before reallocating purchasing budgets. High prices alone do not demonstrate high profitability.

• Track availability over time to distinguish persistent stockouts from temporary catalogue snapshots.

8. Data Quality and Interpretation Limits

The initial preview contains repeated SKUs and mismatches between some URLs and product names. The workflow does not remove duplicates, so repeated records may inflate brand counts and summed exposure. These issues require validation before operational use.

Extracting only the first word after "by" can retain malformed labels such as "StradivariusJacket". A structured extraction method and a verified brand dictionary would improve grouping accuracy. The 27 unavailable entries for one jeans product also require checks for repeated size entries or parsing issues.

Each brand stockout rate represents an unweighted average of product-level rates, rather than the proportion of unavailable sizes across the entire brand. Assigning zero to non-string size values could understate stockouts if availability data is missing.

The catalogue data does not include transactions, stock quantities, customer demand, margins or inventory age. These limitations prevent confirmation of actual revenue loss, high-demand winners, excess inventory or the causes of supply chain inefficiencies.

9. Project Deliverables and Documentation

I documented the data loading and cleaning workflow, brand mapping, stockout feature engineering, product rankings and static brand strategy visualization in a Python notebook. The project scope covers Python analysis and visualization.

Further documentation should record the data source and collection date, currency, skipped-row count, final sample sizes, duplicate-handling rules and validated brand mappings to improve reproducibility.